Vectors in Euclidean Space
Lecture Notes for Multivariable Calculus

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1 Lecture Scope and Goals (5 min)

This lecture is designed as a 2-hour 30-minute introduction to vectors in Euclidean space for students beginning multivariable calculus. The central goals are:

- understand vectors as quantities with magnitude and direction,
- move fluently between geometric pictures and coordinate formulas,
- learn the algebra of vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$,
- connect dot product, cross product, determinants, and matrices to geometry,
- prepare for gradients, tangent planes, linear approximations, and Jacobians.

Suggested Time Allocation

- 25 min: vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$, magnitude, direction, components,
- 25 min: vector addition, scalar multiplication, linear combinations,
- 25 min: dot product, angle, projection,
- 20 min: cross product in $\mathbb{R}^3$,
- 15 min: determinants in dimensions 2 and 3 and geometric meaning,
- 10 min: coordinate changes and area/volume in new coordinates,
- 15 min: matrices as actions on vectors,
- 10 min: integrated examples,
- remaining time: discussion, student problem-solving, and review.

Key Concepts for This Lecture

In this lecture, we will focus on the core ideas of **vectors**, **Euclidean space**, **vector addition**, **scalar multiplication**, **linear combinations**, **dot product**, **cross product**, **determinants**, **coordinate changes**, and **matrices as linear transformations**. These ideas lead directly to the language of **gradients**, **directional derivatives**, **Jacobians**, and geometric thinking in multivariable calculus.

2 Euclidean Space and the Meaning of a Vector (12 min)

The Euclidean space $\mathbb{R}^n$ is the set of all ordered n-tuples

$$(x_1, x_2, \ldots, x_n), \qquad x_i \in \mathbb{R}.$$

For this lecture, the most important cases are:

$$\mathbb{R}^2 = \{(x, y)\}, \qquad \mathbb{R}^3 = \{(x, y, z)\}.$$

2.1 Two Complementary Viewpoints

An element of $\mathbb{R}^n$ can be interpreted in two ways:

- as a **point**, specifying a location,
- as a **vector**, specifying a displacement from the origin.

  For example, the coordinate pair $(3, 2)$ can mean:

- the point located 3 units right and 2 units up, or
- the vector that moves any point by 3 units in the x-direction and 2 units in the y-direction.

2.2 Free Vectors and Position Vectors

A vector is often drawn as an arrow. Its essential information is:

- its length,
- its direction.

The exact starting location does not matter if we only care about displacement. Such an object is a **free vector**. When the tail is placed at the origin, it is called a **position vector**.

2.3 Vectors in Dimension 2

In $\mathbb{R}^2$, a vector

$$\mathbf{v} = (a, b)$$

represents a motion:

- move a units horizontally,
- move b units vertically.

This immediately connects vectors with velocity, force, and displacement in planar motion.

2.4 Vectors in Dimension 3

In $\mathbb{R}^3$, a vector

$$\mathbf{v} = (a, b, c)$$

represents a spatial motion:

- a units in the x-direction,
- b units in the y-direction,
- c units in the z-direction.

This is the natural language for surfaces, tangent planes, fluid flow, and vector fields.

3 Coordinates, Components, and Standard Basis (8 min)

The standard basis vectors are

$$\mathbf{e}_1 = (1, 0, \ldots, 0), \quad \mathbf{e}_2 = (0, 1, 0, \ldots, 0), \quad \ldots, \quad \mathbf{e}_n = (0, \ldots, 0, 1).$$

Every vector $\mathbf{x} = (x_1, \ldots, x_n)$ can be written as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n.$$

3.1 Geometric Meaning of Components

The numbers $x_1, \ldots, x_n$ are not arbitrary labels. They tell us how much of the vector lies in each coordinate direction.

For example, in $\mathbb{R}^2$:

$$(4, -1) = 4\mathbf{e}_1 - \mathbf{e}_2.$$

This says: move 4 units right, then 1 unit down.

In $\mathbb{R}^3$:

$$(2, -3, 5) = 2\mathbf{e}_1 - 3\mathbf{e}_2 + 5\mathbf{e}_3.$$

This is a decomposition into three independent directional actions.

4 Vector Addition, Scalar Multiplication, and Linear Combinations (15 min)

Let

$$\mathbf{u} = (u_1, \ldots, u_n), \qquad \mathbf{v} = (v_1, \ldots, v_n).$$

4.1 Addition

Vector addition is defined componentwise:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \ldots, u_n + v_n).$$

Geometrically, addition combines displacements. If $\mathbf{u}$ moves you first and $\mathbf{v}$ moves you second, the total effect is $\mathbf{u} + \mathbf{v}$.

4.2 Scalar Multiplication

For $c \in \mathbb{R}$,

$$c\mathbf{u} = (cu_1, \ldots, cu_n).$$

Geometrically:

- if $c > 1$, the vector is stretched,
- if $0 < c < 1$, it is compressed,
- if $c < 0$, its direction reverses and its length scales by $|c|$.

4.3 Linear Combinations

An expression of the form

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k$$

is a linear combination.

This is one of the most important ideas in multivariable calculus and linear algebra:

- in $\mathbb{R}^2$, linear combinations of two nonparallel vectors can reach the entire plane,
- in $\mathbb{R}^3$, linear combinations of three suitably independent vectors can reach the entire space.

4.4 Geometric Interpretation in $\mathbb{R}^2$

If

$$\mathbf{v}_1 = (1, 0), \qquad \mathbf{v}_2 = (0, 1),$$

then

$$(x, y) = x\mathbf{v}_1 + y\mathbf{v}_2.$$

This means all points in the plane can be built by combining horizontal and vertical moves.

4.5 Geometric Interpretation in $\mathbb{R}^3$

If

$$\mathbf{v}_1 = (1, 0, 0), \qquad \mathbf{v}_2 = (0, 1, 0), \qquad \mathbf{v}_3 = (0, 0, 1),$$

then

$$(x, y, z) = x\mathbf{v}_1 + y\mathbf{v}_2 + z\mathbf{v}_3.$$

This is the coordinate model of ordinary space.

5 Magnitude, Distance, and Direction (8 min)

5.1 Length of a Vector

The norm of $\mathbf{v} = (v_1, \ldots, v_n)$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}.$$

This is the Euclidean length of the vector.

Examples:

$$\|(3, 4)\| = 5, \qquad \|(2, -1, 2)\| = 3.$$

5.2 Distance Between Points

For points $P = (x_1, \ldots, x_n)$ and $Q = (y_1, \ldots, y_n)$,

$$d(P, Q) = \|Q - P\|.$$

So distance is just the length of the displacement vector from P to Q.

5.3 Unit Vectors

If $\mathbf{v} \neq \mathbf{0}$, then

$$\frac{\mathbf{v}}{\|\mathbf{v}\|}$$

is the unit vector in the direction of $\mathbf{v}$.

Unit vectors separate **direction** from **magnitude**. This is essential in directional derivatives.

6 Dot Product, Angle, and Projection (18 min)

For

$$\mathbf{u} = (u_1, \ldots, u_n), \qquad \mathbf{v} = (v_1, \ldots, v_n),$$

the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n.$$

6.1 Algebraic Role

The dot product combines coordinate information into a single scalar. It measures directional agreement:

- large positive value: vectors point in similar directions,
- zero: vectors are perpendicular,
- negative value: vectors point in opposite general directions.

6.2 Geometric Formula

For nonzero vectors,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta,$$

where θ is the angle between the vectors. Therefore,

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

6.3 Orthogonality

Two vectors are orthogonal if

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

In multivariable calculus, this appears in:

- normal vectors to curves and surfaces,
- gradients perpendicular to level sets,
- optimization constraints.

A Clean Constraint Example (ML/AI Motivation)

Consider the problem

$$\min_{\mathbf{w}\in\mathbb{R}^2} f(\mathbf{w}) = w_1^2 + w_2^2 \quad \text{subject to} \quad w_1 + 2w_2 = 1.$$

We are looking for the vector of smallest length that still satisfies the linear constraint. This is a geometric optimization problem:

- the objective $f(\mathbf{w}) = \|\mathbf{w}\|^2$ measures model size,
- the constraint $w_1 + 2w_2 = 1$ is a line,
- the minimum occurs where the smallest circle centered at the origin touches that line.

The constraint can be written as

$$\mathbf{a} \cdot \mathbf{w} = 1, \qquad \text{where } \mathbf{a} = (1, 2).$$

The line $\mathbf{a} \cdot \mathbf{w} = 1$ has normal vector $\mathbf{a}$, so at the optimal point the vector $\mathbf{w}$ must be parallel to $\mathbf{a}$:

$$\mathbf{w} = \lambda(1, 2).$$

Substitute into the constraint:

$$(1, 2) \cdot \lambda(1, 2) = \lambda(1 + 4) = 5\lambda = 1,$$

so

$$\lambda = \frac{1}{5}, \qquad \mathbf{w}^* = \left(\frac{1}{5}, \frac{2}{5}\right).$$

This is the minimum-norm solution.

Why this matters in machine learning. This same geometry appears constantly in AI and data science:

- in **regularization**, we prefer smaller parameter vectors because they are more stable and often generalize better,
- in **linear models**, we may require a prediction constraint while choosing the smallest weights that satisfy it,
- in **support vector methods**, geometric separation and constrained optimization are built from the same dot-product and normal-vector ideas,
- in modern **machine learning systems**, constrained optimization is used whenever we balance fit, resource limits, or safety constraints.

The key geometric principle is:

At a constrained optimum, the objective and the constraint meet tangentially, so their normals align.

This is the intuition behind Lagrange multipliers.

6.4 Projection

The projection of $\mathbf{u}$ onto a nonzero vector $\mathbf{v}$ is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

This gives the part of $\mathbf{u}$ acting in the direction of $\mathbf{v}$.

6.5 Interpretation as Action

Suppose $\mathbf{v}$ describes a direction of motion. Then $\operatorname{proj}_{\mathbf{v}} \mathbf{u}$ is the effective component of the motion $\mathbf{u}$ that actually contributes along the direction $\mathbf{v}$.

7 Cross Product in $\mathbb{R}^3$ (15 min)

The cross product is defined only in $\mathbb{R}^3$. For

$$\mathbf{u} = (u_1, u_2, u_3), \qquad \mathbf{v} = (v_1, v_2, v_3),$$

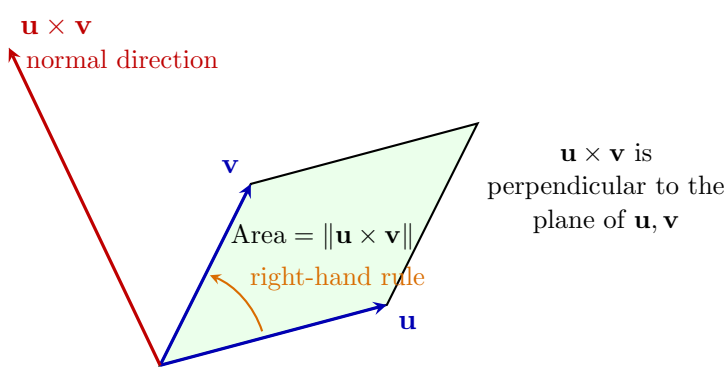
their cross product is

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}.$$

7.1 Geometric Meaning

The vector $\mathbf{u} \times \mathbf{v}$:

- is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$,
- has magnitude
$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta,$$
- points in the direction determined by the right-hand rule.



7.2 Area Interpretation

The quantity

$$\|\mathbf{u} \times \mathbf{v}\|$$

equals the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$.

Thus:

- dot product measures alignment,
- cross product measures oriented perpendicularity and area.

Worked Example: Computing Area in 2D and 3D

In $\mathbb{R}^2$: Let

$$\mathbf{a} = (2, 1), \qquad \mathbf{b} = (1, 3).$$

The area of the parallelogram spanned by these two vectors is computed by the determinant:

$$\text{Area} = \left|\det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\right| = |2 \cdot 3 - 1 \cdot 1| = 5.$$

In $\mathbb{R}^3$: Let

$$\mathbf{u} = (2, 1, 0), \qquad \mathbf{v} = (1, 3, 1).$$

Now the same geometric object is still a parallelogram, but it lies in a plane inside three-dimensional space. Its area is computed by the magnitude of the cross product:

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} 1 \cdot 1 - 0 \cdot 3 \\ 0 \cdot 1 - 2 \cdot 1 \\ 2 \cdot 3 - 1 \cdot 1 \end{pmatrix} = (1, -2, 5).$$

So

$$\text{Area} = \|\mathbf{u} \times \mathbf{v}\| = \sqrt{1^2 + (-2)^2 + 5^2} = \sqrt{30}.$$

What is the difference?

- In **2D**, the area is obtained from a **scalar determinant**. There is only one plane, so no extra normal-direction information is needed.
- In **3D**, two vectors still span a **2D parallelogram**, but it can sit in many different planes. The cross product first produces a **normal vector**, and then its magnitude gives the area.
- So the geometric meaning is the same in both cases: we measure the area of the parallelogram generated by two vectors. The difference is that in $\mathbb{R}^3$ the formula also records the perpendicular direction of that plane.

7.3 Example

Let

$$\mathbf{u} = (1, 2, 0), \qquad \mathbf{v} = (3, 1, 4).$$

Then

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} 2 \cdot 4 - 0 \cdot 1 \\ 0 \cdot 3 - 1 \cdot 4 \\ 1 \cdot 1 - 2 \cdot 3 \end{pmatrix} = (8, -4, -5).$$

This vector is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

8 Determinants in Dimension 2 and 3 (15 min)

Determinants encode signed area, signed volume, orientation, and invertibility.

8.1 Determinant of a 2×2 Matrix

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

8.2 Geometric Meaning in $\mathbb{R}^2$

If the columns of A are vectors

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix},$$

then

$$|\det(A)|$$

is the area of the parallelogram spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$.

The sign of $\det(A)$ records orientation:

- positive: orientation preserved,
- negative: orientation reversed.

8.3 Determinant of a 3×3 Matrix

For

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

the determinant can be computed by cofactor expansion. Along the first row:

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

8.4 Geometric Meaning in $\mathbb{R}^3$

If the columns of A are $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$, then

$$|\det(A)|$$

is the volume of the parallelepiped spanned by those three vectors.

Again, the sign records orientation.

8.5 Scalar Triple Product

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$,

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{pmatrix}.$$

Its absolute value is the volume of the parallelepiped formed by $\mathbf{u}, \mathbf{v}, \mathbf{w}$.

8.6 Why Determinants Matter in Calculus

Determinants appear throughout multivariable calculus:

- Jacobian determinants for change of variables,
- area and volume scaling under linear maps,
- testing whether a linear system has a unique solution,
- orientation of coordinate transformations.

9 Changing Coordinates with Matrices and Computing Area/Volume (10 min)

This section makes the geometric meaning of the determinant explicit.

9.1 A New Coordinate System from New Basis Vectors

In the standard coordinate system of $\mathbb{R}^2$, we use the basis

$$\mathbf{e}_1 = (1, 0), \qquad \mathbf{e}_2 = (0, 1).$$

Suppose instead we choose two new basis vectors

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$

Then every vector written in the new coordinates as

$$\begin{pmatrix} u \\ v \end{pmatrix}_{\text{new}}$$

corresponds to the standard-coordinate vector

$$u\mathbf{v}_1 + v\mathbf{v}_2.$$

If we place the new basis vectors as the columns of a matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then the conversion from new coordinates to standard coordinates is

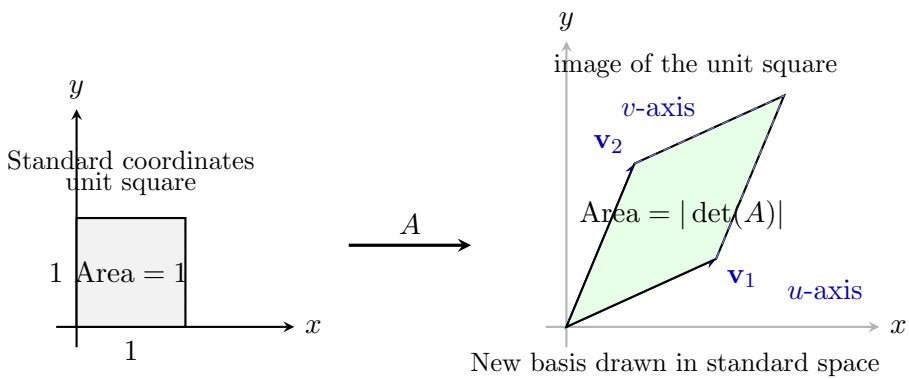
$$\mathbf{x}_{\text{standard}} = A\mathbf{x}_{\text{new}}.$$

9.2 Area in the New Coordinate System

In the new coordinates, the unit square is the set of points

$$0 \le u \le 1, \qquad 0 \le v \le 1.$$

Under the matrix A, that unit square becomes the parallelogram spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$ in standard coordinates.



On the left, the shaded region is the unit square in the (u, v)-coordinates. On the right, after applying A, the same coordinate square is represented in standard (x, y)-space as a parallelogram spanned by the new basis vectors.

Its area is

$$|\det(A)|\,.$$

Therefore:

- one unit of area in the new coordinate system becomes $|\det(A)|$ units of area in the standard system,
- equivalently, the determinant tells us exactly how area scales when coordinates are changed by a matrix.

9.3 Example in $\mathbb{R}^2$

Let

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Then

$$\det(A) = 5.$$

So:

- the unit square in the new coordinates becomes a parallelogram of area 5,
- any region with area S in the new coordinates has area $5S$ in the standard coordinates.

9.4 The Same Idea in $\mathbb{R}^3$

Choose a new basis in $\mathbb{R}^3$:

$$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3.$$

Put these vectors into the columns of

$$A = \begin{pmatrix} | & | & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \\ | & | & | \end{pmatrix}.$$

Then

$$\mathbf{x}_{\text{standard}} = A\mathbf{x}_{\text{new}}.$$

The unit cube in the new coordinates becomes the parallelepiped spanned by $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

Its volume is

$$|\det(A)|.$$

So the determinant is the volume scaling factor for the coordinate change.

9.5 Why This Is the Geometric Meaning of Determinant

The determinant is not just a formula. It answers a geometric question:

When we use a matrix to change coordinates, how much do unit area or unit volume change?

The answer is:

- in $\mathbb{R}^2$, area scales by $|\det(A)|$,
- in $\mathbb{R}^3$, volume scales by $|\det(A)|$.

If $\det(A) < 0$, orientation is reversed during the coordinate change.

9.6 Connection to Change of Variables in Calculus

This is exactly the idea behind Jacobians. If a region is described in new coordinates, then when we measure area or volume in the original coordinates, we multiply by the local scaling factor:

$$|\det(J)|.$$

For linear changes of coordinates, the Jacobian matrix is constant and equals the matrix A itself.

10 Lines, Planes, and Geometry from Vectors (8 min)

10.1 Equation of a Line

A line through a point $\mathbf{r}_0$ with direction $\mathbf{d}$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}, \qquad t \in \mathbb{R}.$$

This says:

- start at $\mathbf{r}_0$,
- move repeatedly along the direction $\mathbf{d}$.

So a line can be viewed as the **trace of motion of a particle**: at time t, the particle is at position $\mathbf{r}(t)$.

Exercise (with solution)

For each line below:

- identify a direction vector,
- identify one point on the line,
- rewrite the line in the unified vector-parametric form

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{d}.$$

(a) A line in the plane:

$$2x - y = 3.$$

Solution. One point on the line is $(2, 1)$, since $2(2) - 1 = 3$.
To find a direction vector, use the fact that the normal vector of the line is

$$\mathbf{n} = (2, -1).$$

A direction vector must be perpendicular to $\mathbf{n}$, so one convenient choice is

$$\mathbf{d} = (1, 2),$$

because

$$(2, -1) \cdot (1, 2) = 2 - 2 = 0.$$

Hence the line can be written as

$$\mathbf{r}(t) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

In coordinates:

$$x = 2 + t, \qquad y = 1 + 2t.$$

(b) A line in $\mathbb{R}^3$ given in symmetric form:

$$\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z}{4}.$$

Solution. Set the common value equal to t. Then

$$\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z}{4} = t.$$

This gives

$$x = 1 + 2t, \qquad y = -3 - t, \qquad z = 4t.$$

So one point on the line is

$$(1, -3, 0),$$

and a direction vector is

$$(2, -1, 4).$$

Therefore the unified vector-parametric form is

$$\mathbf{r}(t) = \begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}, \qquad t \in \mathbb{R}.$$

Key idea. Different algebraic descriptions may look different, but geometrically they describe the same object: a particle starting at some point and moving forever in a fixed direction.

10.2 Equation of a Plane in $\mathbb{R}^3$

A plane through a point $\mathbf{r}_0$ with normal vector $\mathbf{n}$ satisfies

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0.$$

The normal vector controls the orientation of the plane.

10.3 Cross Product and Planes

If two nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$ lie in a plane, then

$$\mathbf{u} \times \mathbf{v}$$

is a normal vector to that plane. This is one of the most common practical uses of the cross product.

11 Matrices as Actions on Vectors (15 min)

Students often first see matrices as rectangular arrays of numbers. For multivariable calculus, the more useful viewpoint is:

A matrix represents a rule that sends vectors to vectors.

11.1 Matrix-Vector Multiplication

If

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

then

$$A\mathbf{x} = \begin{pmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{pmatrix}.$$

This means the output coordinates are built from linear combinations of the input coordinates.

11.2 Column Interpretation

Write

$$A = \begin{pmatrix} | & | \\ \mathbf{c}_1 & \mathbf{c}_2 \\ | & | \end{pmatrix}.$$

Then

$$A \begin{pmatrix} x \\ y \end{pmatrix} = x\mathbf{c}_1 + y\mathbf{c}_2.$$

This is the cleanest geometric interpretation:

- the first column is where $\mathbf{e}_1$ goes,
- the second column is where $\mathbf{e}_2$ goes.

  So a matrix is completely determined by what it does to the basis vectors.

11.3 Action in $\mathbb{R}^2$

If

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

then

$$A(x, y) = (2x, 3y).$$

This stretches space by a factor of 2 in the x-direction and by a factor of 3 in the y-direction.

If

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

then

$$R(x, y) = (-y, x),$$

which rotates vectors by 90° counterclockwise.

11.4 Action in $\mathbb{R}^3$

If

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

then

$$A(x, y, z) = (x, y, 2z).$$

This leaves the horizontal coordinates fixed and stretches in the vertical direction.

11.5 Meaning of Matrix Addition

If A and B are matrices of the same size, then

$$(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}.$$

So matrix addition means:

- apply two linear actions to the same vector,
- then add the resulting vectors.

It combines two linear effects into one.

11.6 Meaning of Scalar Multiplication of Matrices

For a scalar c,

$$(cA)\mathbf{x} = c(A\mathbf{x}).$$

This rescales the output of the transformation.

11.7 Meaning of Matrix Multiplication

If A and B have compatible sizes, then

$$(AB)\mathbf{x} = A(B\mathbf{x}).$$

This is the most important structural fact:

- first apply B,
- then apply A.

So matrix multiplication represents **composition of actions**.

11.8 Why Order Matters

Usually,

$$AB \neq BA.$$

Geometrically, doing one action and then another often differs from reversing the order. For example, stretching then rotating is generally not the same as rotating then stretching.

Example: A Stretch and a Rotation

Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Here, A stretches horizontally, while B rotates vectors by 90° counterclockwise.
Now compute both orders:

$$A\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad B\mathbf{x} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

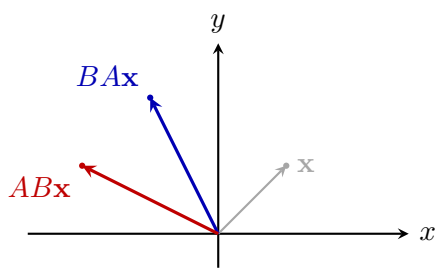
So

$$BA\mathbf{x} = B(A\mathbf{x}) = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \qquad AB\mathbf{x} = A(B\mathbf{x}) = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Since

$$BA\mathbf{x} \neq AB\mathbf{x},$$

the order of the two actions matters.



Geometric message. The same starting vector produces different outputs because the first action changes the direction or shape of the vector before the second action is applied.

Think about this. Although

$$AB \neq BA,$$

we still have

$$\det(AB) = \det(BA).$$

How can you explain this geometrically?
Hint: both compositions scale area by the product of the two individual area-scaling factors. One order may send a square to a different *shape* than the other, but the total area scale is the same:

$$|\det(A)|\,|\det(B)|\,.$$

So the final pictures can look different while the total area expansion factor remains identical.

11.9 Determinant as Area/Volume Scaling

If A is a linear transformation:

- in $\mathbb{R}^2$, $|\det(A)|$ is the factor by which areas scale,
- in $\mathbb{R}^3$, $|\det(A)|$ is the factor by which volumes scale.

Therefore, determinants measure how strongly a matrix expands, contracts, or collapses space.

11.10 Invertibility

If

$$\det(A) \neq 0,$$

then the matrix is invertible, meaning the transformation can be undone.

If

$$\det(A) = 0,$$

then the matrix collapses dimension:

- in $\mathbb{R}^2$, a region may collapse to a line or point,
- in $\mathbb{R}^3$, a volume may collapse to a plane, line, or point.

12 Connection to Multivariable Calculus (4 min)

These topics are not separate preliminaries. They are the language of multivariable calculus.

12.1 Gradients

For a scalar function $f(x_1, \ldots, x_n)$,

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right).$$

This is a vector of partial derivatives.

12.2 Directional Derivatives

If $\mathbf{u}$ is a unit vector, then

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.$$

The dot product turns the gradient into the rate of change in the chosen direction.

12.3 Derivative Matrix

For a vector-valued map

$$F : \mathbb{R}^n \to \mathbb{R}^m,$$

the derivative is represented by a matrix, the Jacobian matrix. Near a point, the function acts approximately like a linear transformation. That is why matrices and vectors are foundational.

13 Worked Examples (6 min)

Example 1: Addition as Combined Motion in $\mathbb{R}^2$

Let

$$\mathbf{u} = (3, 1), \qquad \mathbf{v} = (-2, 4).$$

Then

$$\mathbf{u} + \mathbf{v} = (1, 5).$$

Interpretation:

- first move right 3 and up 1,
- then move left 2 and up 4,
- net result: right 1 and up 5.

Example 2: Projection in $\mathbb{R}^2$

Let

$$\mathbf{u} = (3, 4), \qquad \mathbf{v} = (1, 1).$$

Then

$$\mathbf{u} \cdot \mathbf{v} = 7, \qquad \mathbf{v} \cdot \mathbf{v} = 2,$$

so

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{7}{2}(1, 1) = \left(\frac{7}{2}, \frac{7}{2}\right).$$

Example 3: Area from a Determinant in $\mathbb{R}^2$

Let

$$\mathbf{v}_1 = (2, 1), \qquad \mathbf{v}_2 = (1, 3).$$

Using the matrix with these as columns:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix},$$

we get

$$\det(A) = 2 \cdot 3 - 1 \cdot 1 = 5.$$

Therefore, the parallelogram spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$ has area 5.

Example 4: Volume from a Determinant in $\mathbb{R}^3$

Let

$$\mathbf{u} = (1, 0, 0), \qquad \mathbf{v} = (1, 2, 0), \qquad \mathbf{w} = (1, 2, 3).$$

Then

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix} = 6.$$

So the parallelepiped spanned by these vectors has volume 6.

Example 5: Matrix as a Transformation

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Then

$$A(x, y) = (x + 2y, y).$$

This is a shear:

- horizontal position changes depending on height,
- vertical position stays unchanged.

14 Student Problems: Computational Practice (7 min)

1. In $\mathbb{R}^2$, draw the vectors $(2, 1)$, $(-1, 3)$, and their sum. Describe the combined motion in words.
2. Compute the norm and a unit vector in the direction of $(6, 8)$.
3. Find the angle between $(1, 0)$ and $(1, \sqrt{3})$.
4. Determine whether $(2, -1, 3)$ and $(1, 4, 0)$ are orthogonal.
5. Compute the projection of $(4, 1)$ onto $(1, 2)$.
6. Find the cross product of $(1, 2, 3)$ and $(0, 1, 4)$.
7. Verify directly that your cross product in the previous problem is perpendicular to both original vectors.
8. Write the equation of the line through $(1, -2, 3)$ parallel to $(2, 1, -1)$.
9. Write the equation of the plane through $(1, 0, 2)$ with normal vector $(3, -1, 4)$.
10. Compute the determinant of
$$\begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$$
and interpret the result geometrically.
11. Compute the determinant of
$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$
12. Let
$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$
Compute $A(1, 0)$, $A(0, 1)$, and $A(2, -1)$, and explain what the matrix does to the plane.

15 Student Problems: Construction and Thinking (5 min)

These work well for discussion, board work, or short group activities.

1. In $\mathbb{R}^2$, construct two different nonzero vectors whose sum is $(4, 3)$. How many such pairs exist?
2. In $\mathbb{R}^2$, find all vectors orthogonal to $(2, -5)$. What geometric object do they form?
3. In $\mathbb{R}^2$, can two different vectors have the same length and the same projection onto the x-axis? Give examples and explain.
4. In $\mathbb{R}^3$, find a nonzero vector orthogonal to both $(1, 1, 0)$ and $(1, 0, 1)$ without using the cross product formula first. Then verify your result with the cross product.
5. In $\mathbb{R}^3$, give three vectors that lie in the same plane. Explain how you can detect coplanarity using a determinant.

6. Find two vectors in $\mathbb{R}^2$ that span the same parallelogram area as $(3, 0)$ and $(0, 2)$ but point in different directions.
7. Give an example of a 2×2 matrix with negative determinant. Describe what its action does to orientation.
8. Give an example of a nonzero 2×2 matrix with determinant 0. Describe the geometric collapse it causes.
9. Find a 2×2 matrix that doubles all areas but does not preserve lengths.
10. Find a 3×3 diagonal matrix that preserves volume but changes shape.
11. Consider two matrices: one rotates the plane by $90°$, the other stretches only in the x-direction. Compute both possible products and explain geometrically why the results differ.
12. Suppose a matrix sends both basis vectors in $\mathbb{R}^2$ to the same vector. What must its determinant be? What happens to every square under this map?

16 Short Concept Checks for Class (3 min)

These can be used during the lecture to force interpretation, not just calculation.

1. If $\mathbf{u} \cdot \mathbf{v} > 0$, what does that tell you about the angle between the vectors?
2. If $\det(A) = 0$ for a 2×2 matrix, what can you say about its two column vectors?
3. Why does a cross product in $\mathbb{R}^3$ naturally produce a vector, while a determinant in $\mathbb{R}^2$ produces a number?
4. What does the second column of a matrix tell you geometrically?
5. If a matrix has determinant -1, what does that say about area (or volume) and orientation?

17 Summary (3 min)

The key structural ideas are:

- vectors represent displacement, direction, and magnitude,
- addition and scalar multiplication describe how motions combine,
- the dot product measures alignment and gives angles and projections,
- the cross product in $\mathbb{R}^3$ produces a perpendicular vector and measures area,
- determinants measure signed area in $\mathbb{R}^2$ and signed volume in $\mathbb{R}^3$,
- matrices represent linear actions on vectors,
- matrix multiplication represents composition of actions,
- determinants measure area/volume scaling under matrices.

These ideas form the geometric language behind gradients, Jacobians, tangent approximations, and change of variables in multivariable calculus.